# R.V.S SCHOOL OF ENGINEERING AND TECHNOLOGY
# DINDIGUL

## PROJECT TITLE : SWIFT SHIP TRACKER


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# Phase 1: Requirement Analysis &Planning

“Customers often face delays and confusion while booking, tracking, and managing parcel deliveries because they need to call support or use multiple apps. A conversational AI agent can simplify this by allowing customers to book parcels, get live tracking, and manage deliveries in one place.”

## Key Requirements:

• Requirement gathering from end users and delivery agents.
• Defining system scope and user personas (Customer, Agent, Admin).
• Designing the data model with relationships between Parcel, Sender, Receiver, and Delivery.
• Identifying use cases such as parcel booking, tracking, and delivery update automation.
• Selecting tools: Salesforce Flows, Prompt Builder, and AgentForce AI.

## Defining Project Scope & Objectives

### Project Scope:

- Build a scalable CRM solution for parcel delivery businesses to manage customer bookings, shipments, and delivery operations efficiently.
- Support the end-to-end parcel lifecycle — from parcel booking → dispatch → tracking → delivery confirmation.

- Include comprehensive data model design, automation (Flows, Apex, Email Alerts), AI-powered prompt builder, and reporting dashboards for
- effective management. Provide role-based access for admins, delivery agents, and customers through Salesforce profiles, permission sets, and Experience
- Cloud.
  Ensure real-time parcel visibility for customers and simplified operational tracking for agents through AI-integrated conversational support.

## Objectives Summary:

- Streamline parcel operations by automating booking, tracking, and delivery notifications to reduce manual updates.
- Improve customer satisfaction with transparent, real-time shipment status and self-service tracking options.
- Enhance agent efficiency through centralized dashboards and guided Flows for quick status updates.
- Enable business insights with reports and dashboards showing parcel trends, delivery performance, and agent activity.
- Ensure security and reliability by implementing Salesforce role hierarchies, sharing rules, and audit-based data access.

## Users Involved:

- Customers / Senders – End users who book parcels, track deliveries, and receive real-time updates through the portal or AI agent.
- Delivery Agents – Manage assigned parcels, update delivery status, and ensure timely handover to customers.

- Customer Support Team – Handle delivery-related issues, respond to customer queries, and ensure smooth communication.
- Admin Team – Configure the Salesforce environment, manage user roles, security settings, and monitor operational performance.
- System Integrator / Developer – Responsible for implementing automation, AI agent setup, and backend integrations.

## Key Functional Needs:

- Seamless parcel booking and tracking interface for customers through Salesforce Experience Site or agent assistance.
- Ability for agents to update parcel status (Booked → In Transit → Out for Delivery → Delivered) in real time.
- Automated notifications and email alerts to customers on every delivery stage update.
- Support for Sender and Receiver details management with address, contact, and email fields for communication.
- AI Agent integration (via Prompt Builder) for real-time parcel status queries and automated assistance.
- Reports and dashboards to monitor parcel flow, delivery performance, and agent activity for admins.
- Security controls with role-based access, profiles, and permission sets for different user levels.

## Tools Used:

- Salesforce Developer Org – For configuration, development, and testing.
- Prompt Builder (AgentForce AI) – To create intelligent parcel tracking prompts and responses.
- Salesforce Flow Builder – To automate booking, tracking, and delivery updates.
- Google Sheets / Docs – For initial data planning, requirement documentation, and system mapping.

## Identifying Key Salesforce Features & Tool Required

## Salesforce Features Planned:

● Custom Objects:
Parcel__c, Delivery__c, Sender__c, Receiver__c
● Standard Objects:
Account, Contact, Case, Task, EmailMessage, User
● Automations:
Record-Triggered Flows, Auto-Launched Flows, Approval Processes, Scheduled Flows
● Apex:
Apex Classes for automated parcel operations (e.g., Lead-to-Parcel mapping, delivery notifications), Batch Apex for periodic updates or notifications
● UI:
Lightning App Page (SwiftShip Tracker App), Dynamic Forms for Parcel Details, Lightning Record Pages, and Experience Cloud for Customer Access

● AgentForce AI & Prompt Builder:
Integrated AI agent trained to fetch parcel data based on Parcel ID and respond to tracking or delivery status queries
● Email Services:
Custom Email Alerts and Templates for booking confirmations, delivery updates, and acknowledgments
● Security:
Profiles, Permission Sets, Role Hierarchies, Field-Level Security, and Sharing Rules to ensure controlled access for Admin, Agent, and Customer roles

## Designing Data Model and Security Model

## Data Model Includes:

- Parcel__c (Custom Object)
- Stores core parcel details such as Parcel ID, Status, Weight, and Estimated Delivery Date.
- Linked to Sender__c and Receiver__c objects to fetch sender and receiver details.
- Connected with Delivery__c object to track real-time parcel location and status.
- Delivery__c (Custom Object)
- Stores delivery-specific information such as current location, estimated delivery date, and linked parcel.
- Connected with Sender__c and Parcel__c for relationship-based tracking.
- Used by agents to update and monitor delivery status during the parcel lifecycle.

- Sender__c (Custom Object)
- Captures sender-related information such as address, contact number, and email.
- Linked to multiple parcels to track sender history and associated deliveries.
- Used for sending booking confirmations and delivery updates via automated emails.
- Receiver__c (Custom Object)
- Stores receiver details such as address, contact, and email for communication and delivery confirmation.
- Linked to Sender__c and Parcel__c to associate both ends of the shipment process.
- Used by agents for delivery reference and automated email notifications upon completion.

## Security Model Design:

● Role Hierarchy:
Administrator → Delivery Manager → Agent → Customer
● Profiles:
Admin Profile: Full access to all objects and records.
Delivery Agent Profile: Read/Edit access to Parcel, Delivery, Sender, and Receiver objects.
Customer Profile: Read-only access to their own parcel details via Experience Cloud.
Support Staff Profile: Access to Cases and Parcel details for issue resolution.
● Permission Sets:
Used to grant additional privileges (e.g., “Update Delivery Status,” “Access Agent Console”).

- Record-Level Security:
  Sharing Rules applied for Parcel__c and Delivery__c to allow controlled access for agents and managers.
  Customers can only view records linked to their own Sender or Receiver details.
- Field-Level Security:
  Sensitive fields like Sender Email, Receiver Contact, and Location Coordinates restricted to Admin and Agent profiles only.

# Phase 2: Salesforce Development –Backend & Configurations

This phase focused on developing the core logic of SwiftShip Tracker. Custom objects were created for Parcel, Sender, Receiver, and Delivery with relationships. Automations were built using Flows for parcel status updates and approval processes.

## Milestone 1- Salesforce Account

## Introduction:

Are you new to Salesforce? Not sure exactly what it is, or how to use it? Don’t know where you should start on your learning journey? If you’ve answered yes to any of these questions, then you’re in the right place. This module is for you.

Welcome to Salesforce! Salesforce is game-changing technology, with a host of productivity-boosting features, that will help you sell smarter and faster. As you work toward your badge for this module, we'll take you through these features and answer the question, "What is Salesforce, anyway?".

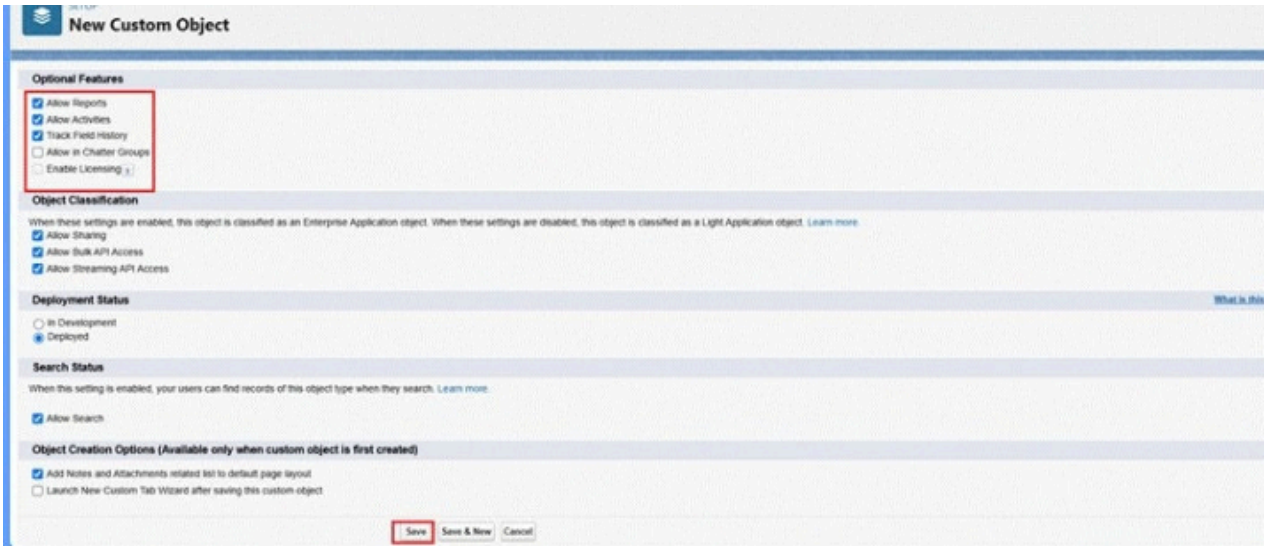
## What Is Salesforce?

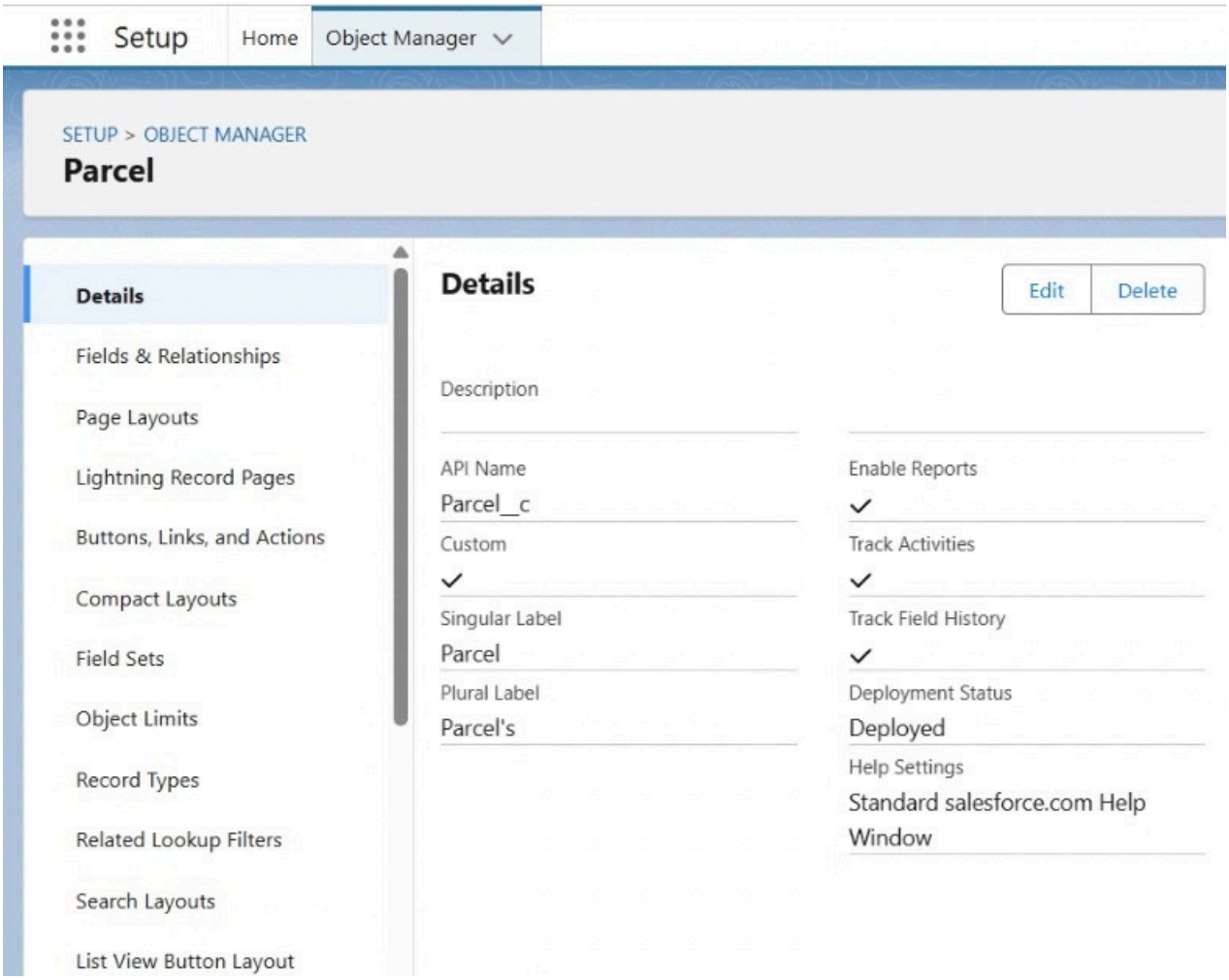
Salesforce is your customer success platform, designed to help you sell, service, market, analyse, and connect with your customers.

## Milestone 2- Objects Creation

To create an object:

1. From the setup page
2.Click on Object Manager
3. Click on Create >> Click on Custom Object.
4. Enter the label name as Parcel
5. Enter Plural label name as Parcel's
6. Enter Record Name as Parcel Name
7. Select Data Type as Text: Parcel Name
8. Select Allow reports.
9. Select Allow search.
10. Allow Track Field History
11. Click on Save and New


New Custom Object
Optional Features
Allow Reports
Allow Activities
Track Field History
Allow in Chatter Groups
Enable Licensing
Object Classification
When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. Learn more.
Allow Sharing
Allow Bulk API Access
Allow Streaming API Access
Deployment Status
In Development
Deployed
Search Status
When this setting is enabled, your users can find records of this object type when they search. Learn more.
Allow Search
Object Creation Options (Available only when custom object is first created)
Add Notes and Attachments related list to default page layout
Launch New Custom Tab Wizard after saving this custom object
Save
Save & New
Cancel

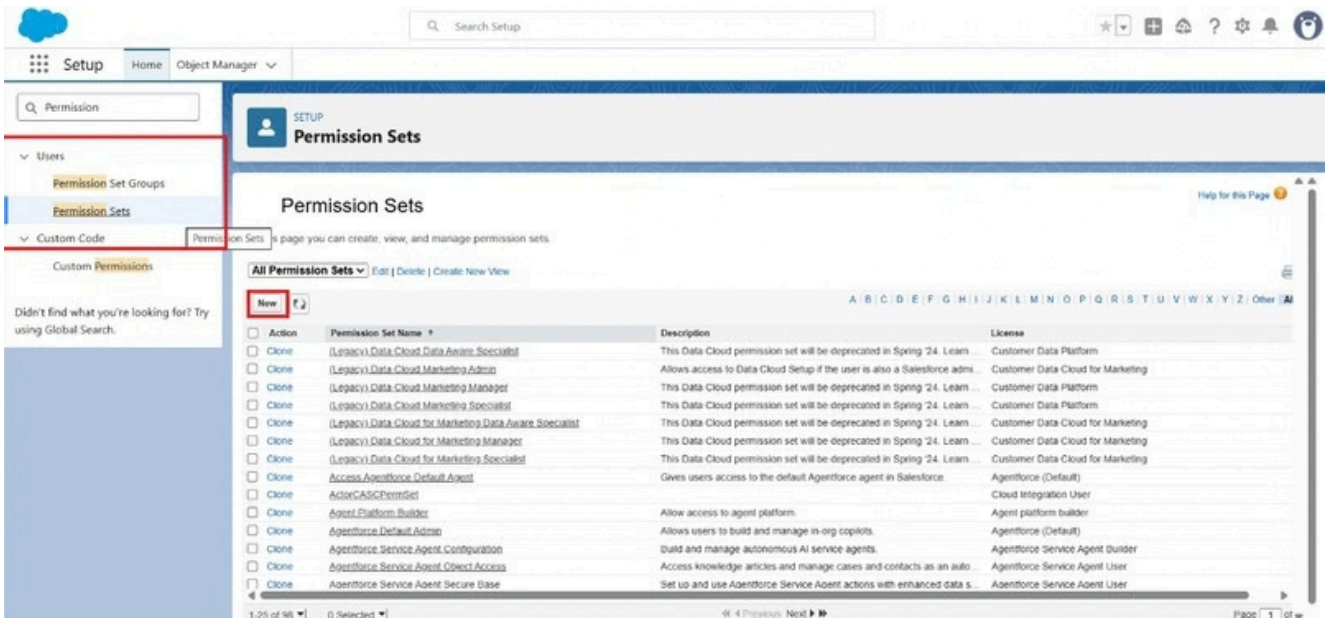

Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Parcel
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Details
Edit
Delete
Description
API Name
Parcel__c
Custom
Singular Label
Parcel
Plural Label
Parcel's
Enable Reports
Track Activities
Track Field History
Deployment Status
Deployed
Help Settings
Standard salesforce.com Help Window

# Phase 3: UI/UX Development & Customization

To enable agents to efficiently manage parcel operations and ensure secure access control using permission sets and AI-driven agent experiences built with AgentForce.

## Milestone - 1 Creating Permission Sets

1.In the Quick Find box (left side), type Permission Sets.
2.Click Permission Sets under Users.



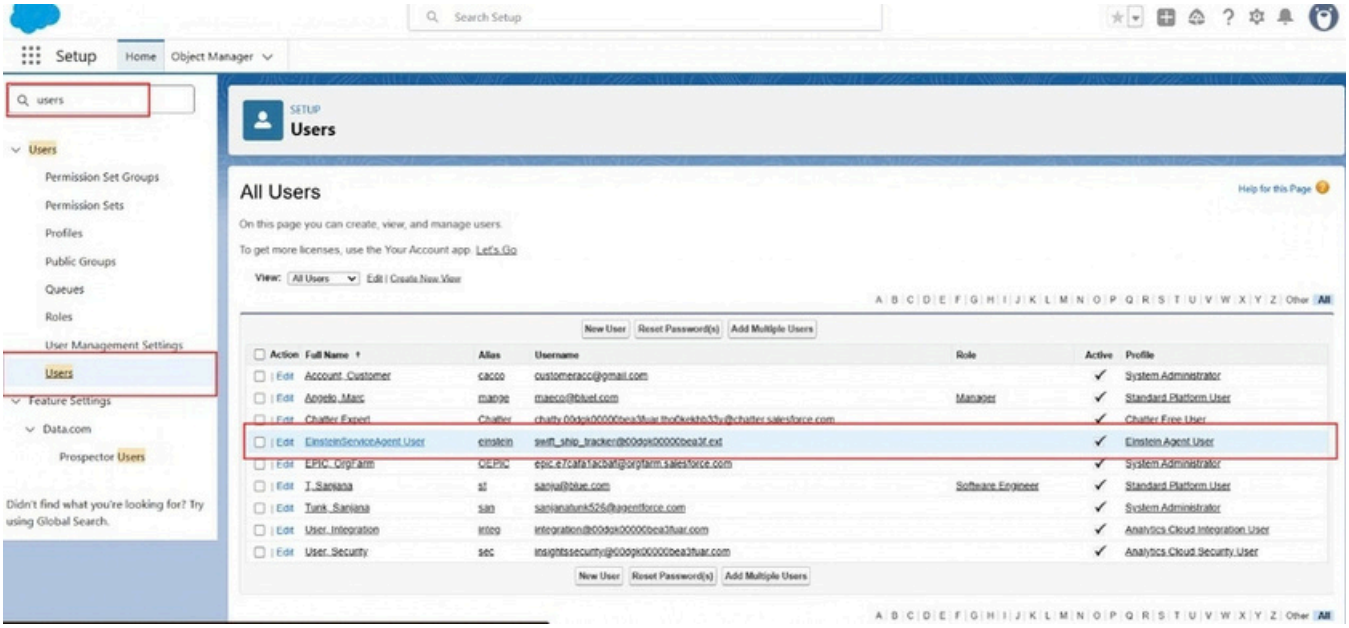

1.Click the New button.
2.Enter the following details:

- Label: Swift Ship
- API Name: auto-fills
- License: Choose Einstein Agent.

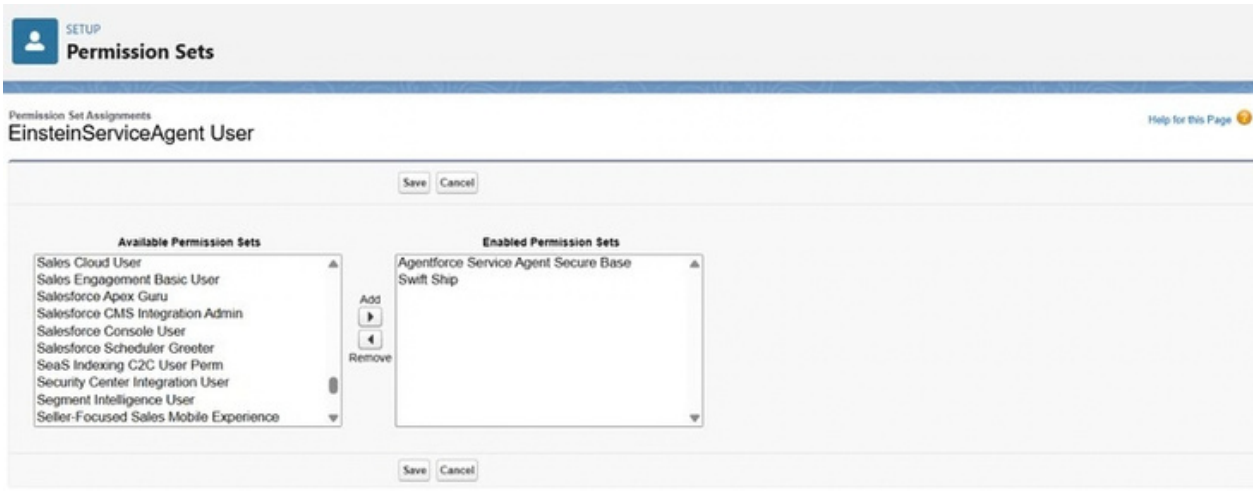
3.Click Save.

# Milestone - 2 Assigning Permissions to your Agent user

1.In the Quick Find box on the left, type Users.
2.Click Users under the Users section.
3.Open the Agent User Record
From the list of users, find the Einstein Service Agent user to represent your Agent (e.g., “SwiftShip Agent”).
Click on the Full Name of that user to open the record.



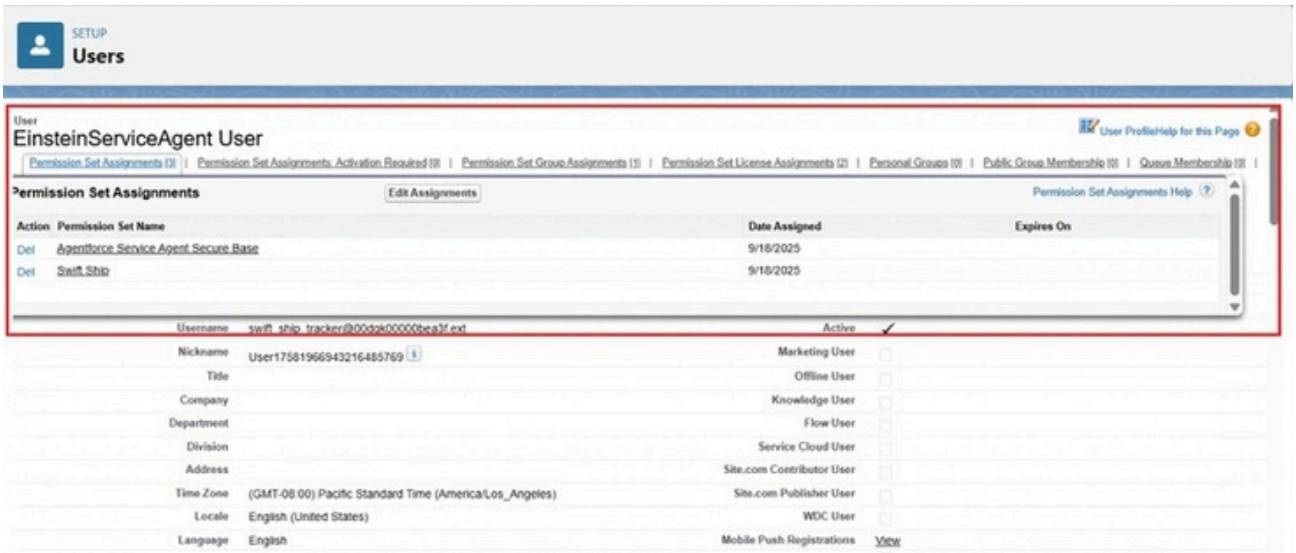

4.Scroll down to the section named Permission Set Assignments.
5.Click the Edit Assignments button.




1.On the left side (Available Permission Sets), find and select your permission set:

- SwiftShip

Click the Add → button to move it to the right-side list (Enabled Permission Sets).
Click Save.

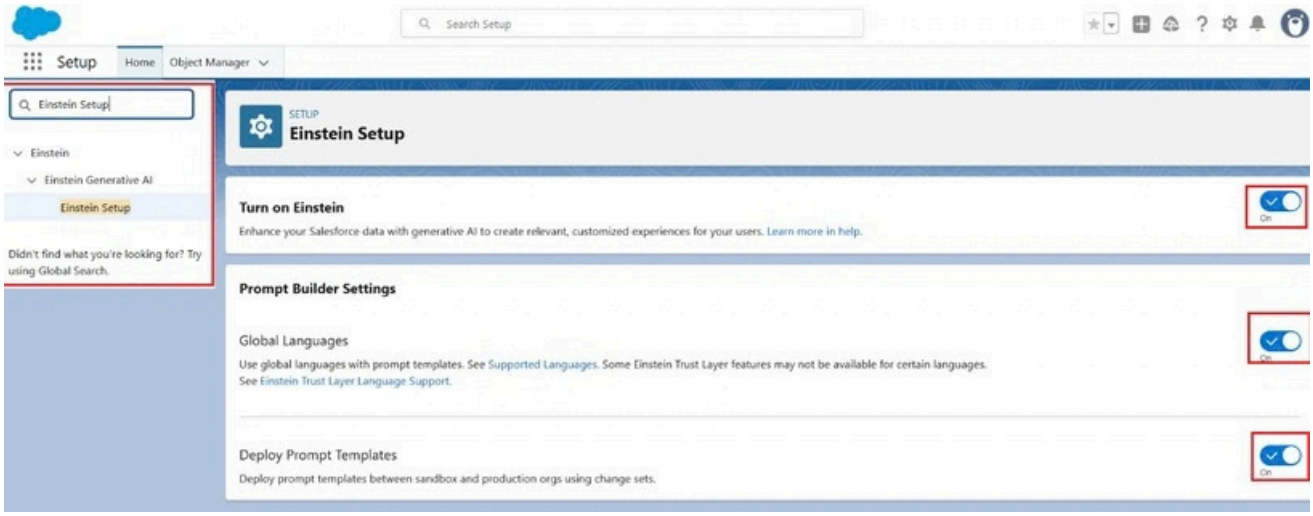



## Milestone -3 Org Setup for AgentForce

1.Go to Setup → Quick Find → Einstein Setup.
2.Open Einstein Setup → click Get Started if prompted.

Toggle On these key services:

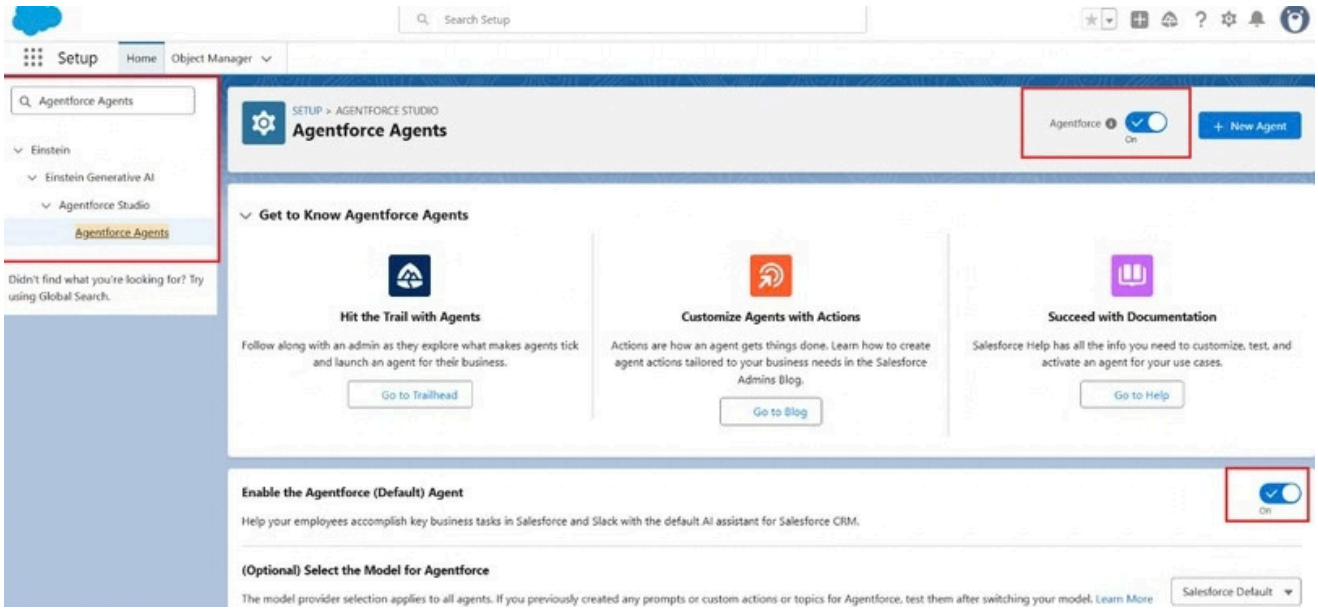
- Turn On Einstein
- Global Languages
- Deploy Prompt Template




3. Refresh your browser page.
4. Go to Setup → Quick Find → AgentForce Agents.

Toggle On these key services:

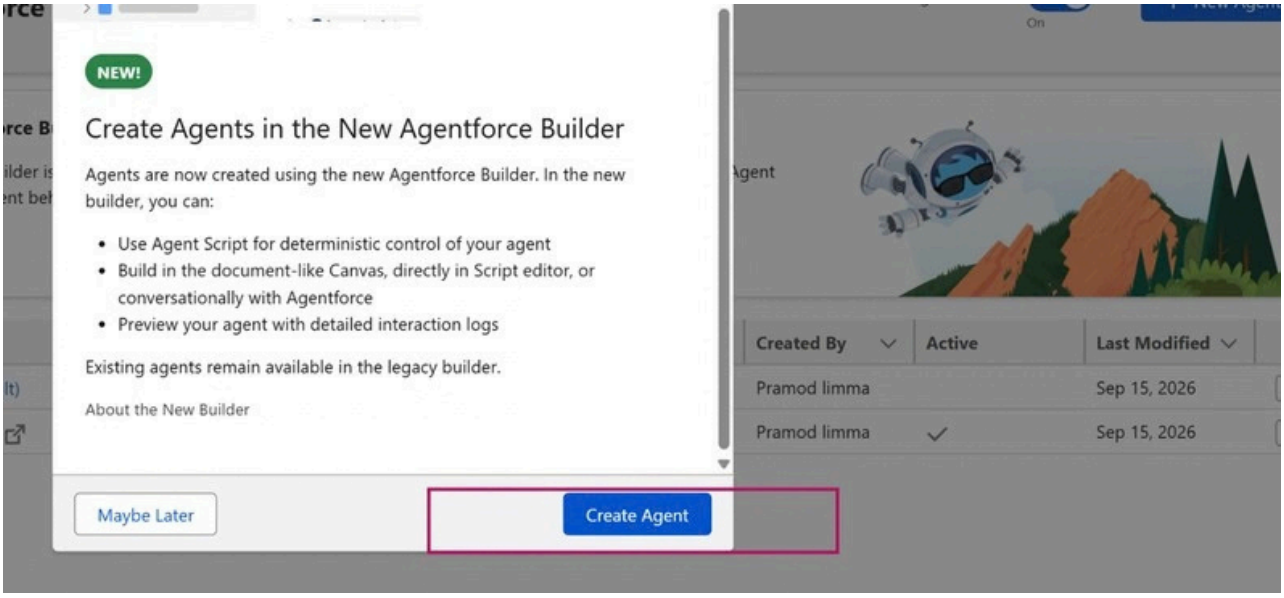
- Agentforce
- Enable Agentforce Default Agent



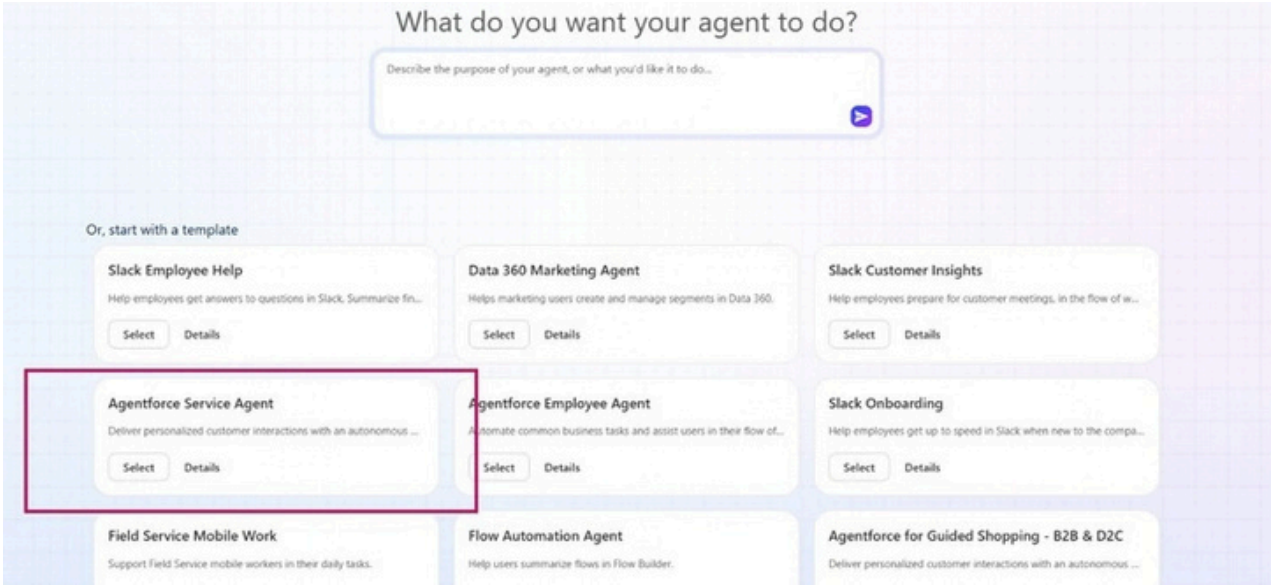
# Milestone - 4 Building The Agent

1.In Setup, search for Agentforce Agents in Quick Find.

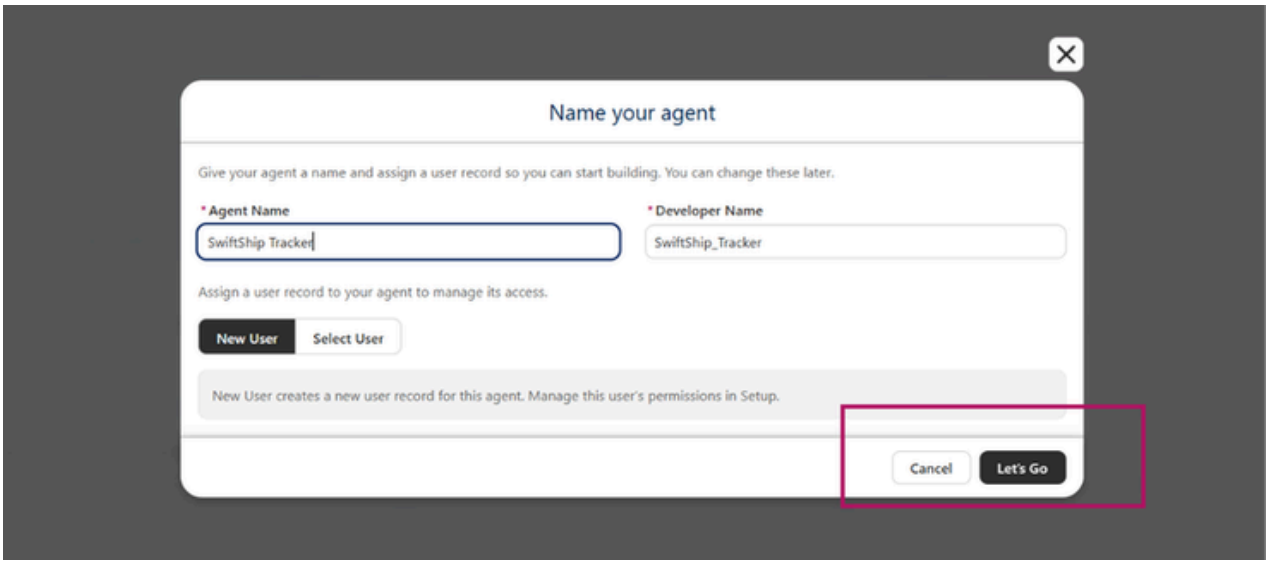
2.Click Agentforce Agents, then click on the +New Agent, Then click Create Agent




Then select Agentforce Service agent



Then create the agent
Label: SwiftShip Tracker
Developer Name : Auto fills



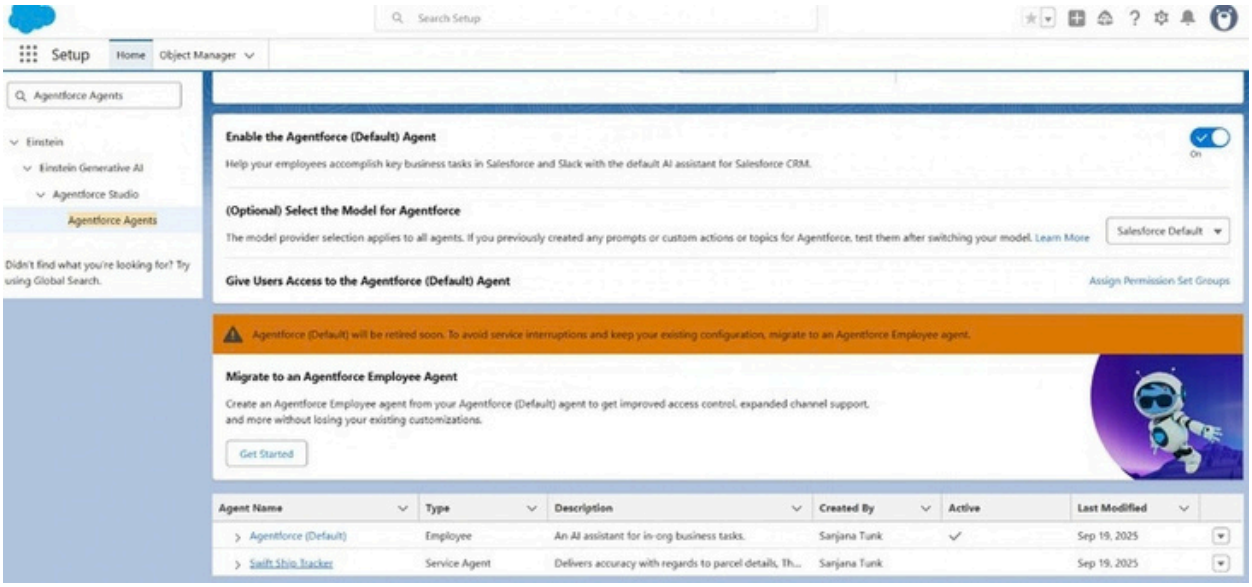

## Phase 4 : Data Migration, Testing & Security

To validate the functionality and reliability of the Parcel Tracking Agent created in Agent Builder. This phase ensures that the agent correctly recognizes topics, triggers the assigned Flow (SwiftShip Tracker), retrieves or updates parcel data, and performs all intended actions without access or permission issues. The goal is to confirm that the AI-driven agent operates seamlessly within Salesforce and provides accurate, real-time responses to parcel management queries.

# Verify Agent

1.Goto Setup →Agent Builder.
2.Open the Agents list and confirm that your agent Name: Parcel Tracking Agent, appears.
Check details:

- Description: “Helps customers book, track, and manage parcels.”
- Type: Service Agent
- Default Agent: Confirm if toggled ON (so it responds automatically).

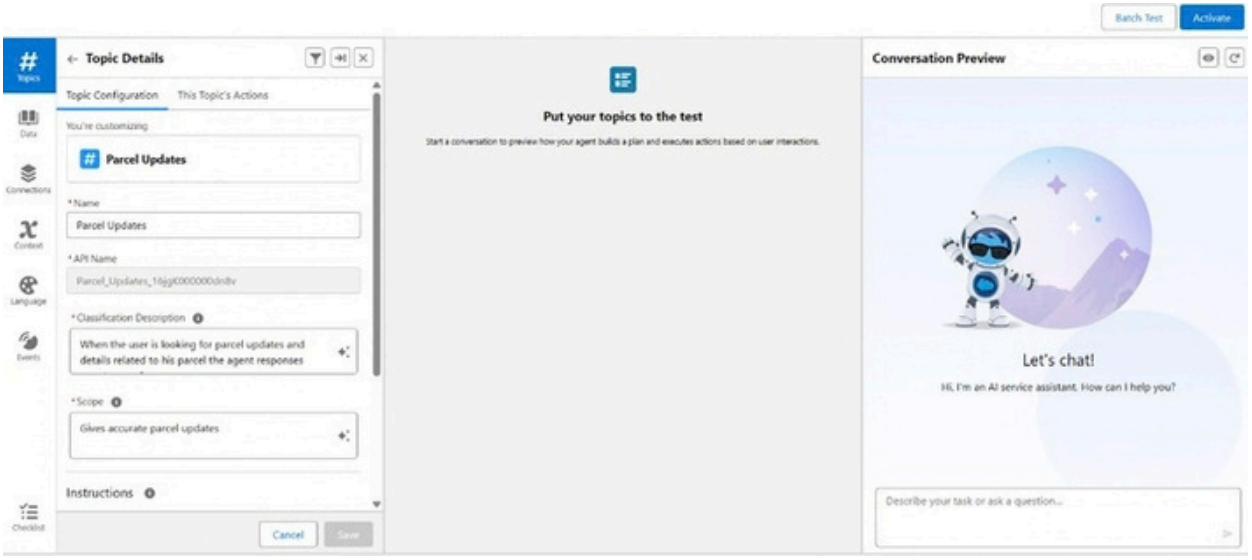



# Validate Assigned Topic

1.In Agent Builder → Topics, confirm that the Topic you created (e.g., Parcel Tracking or SwiftShip Tracker) is listed.
Open the topic to verify:

- It contains a prompt or description related to parcel management.
- The Action/Flow (SwiftShip Tracker) is linked to this topic.

2.Click Save and ensure the topic is Active.

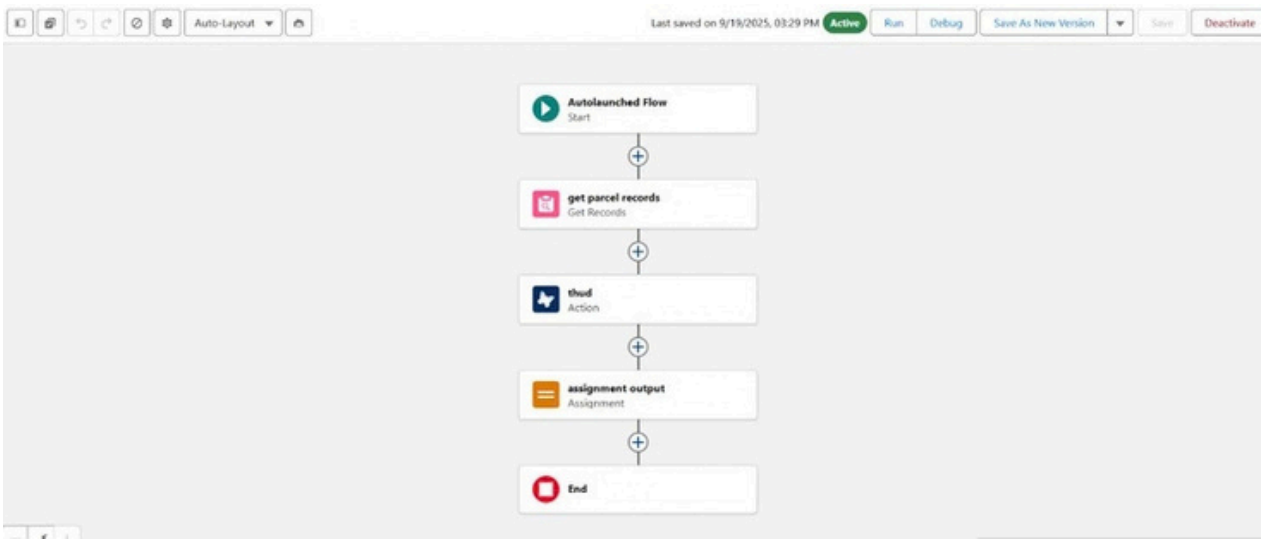



## Check the Flow Association

1.Go to Setup → Flows.
2.Locate the Flow named SwiftShip Tracker.
3.Confirm:

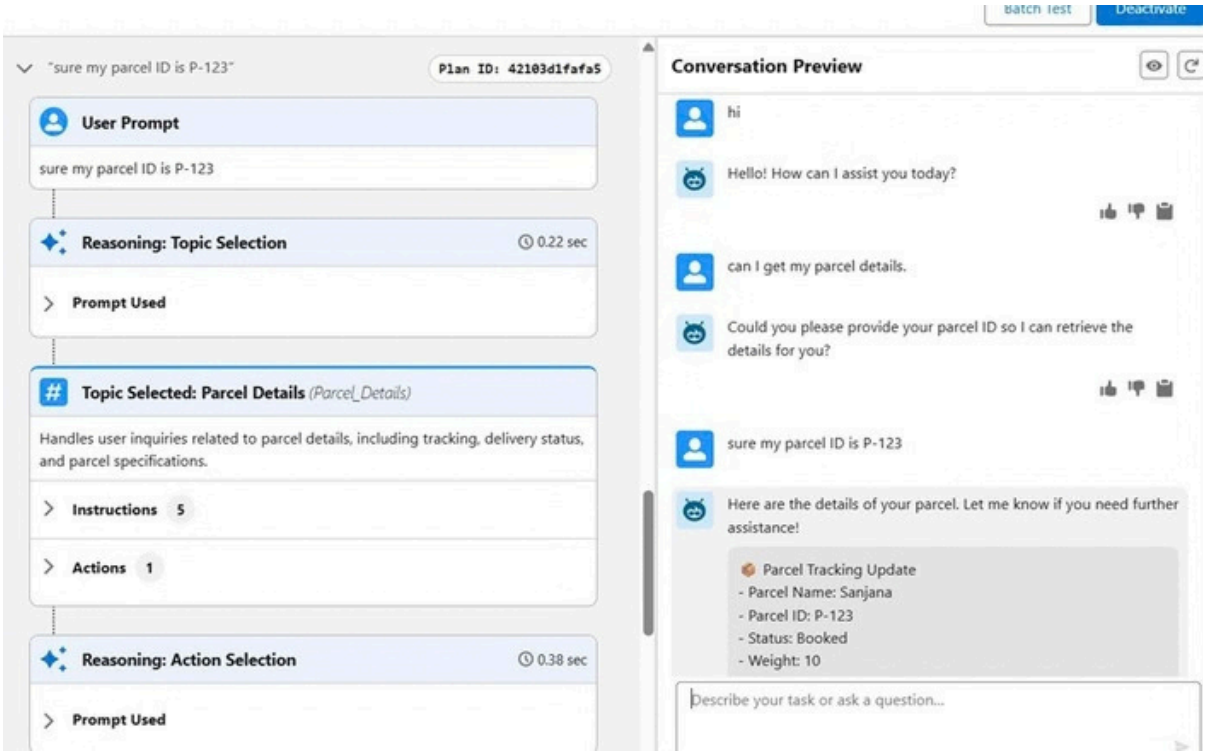
- The Flow is Activated.
- The Flow includes screens or actions that handle:
    1.Parcel ID input
    2.Parcel details retrieval
    3.Status update or confirmation message

4.Open the Flow's Advanced Settings → ensure Run in System Context (Without Sharing) if data access is restricted.



## Activate and Test the Agent

1.Return to Setup → Agent Builder → Parcel Tracking Agent .
2.Click Activate (if not already activated).
3.Once active, test using Preview / Try it Out mode:
Type a query such as:
“Track parcel ID P12345” or
“Update delivery status for parcel P67890 to Delivered.”
4.Observe the agent’s response.

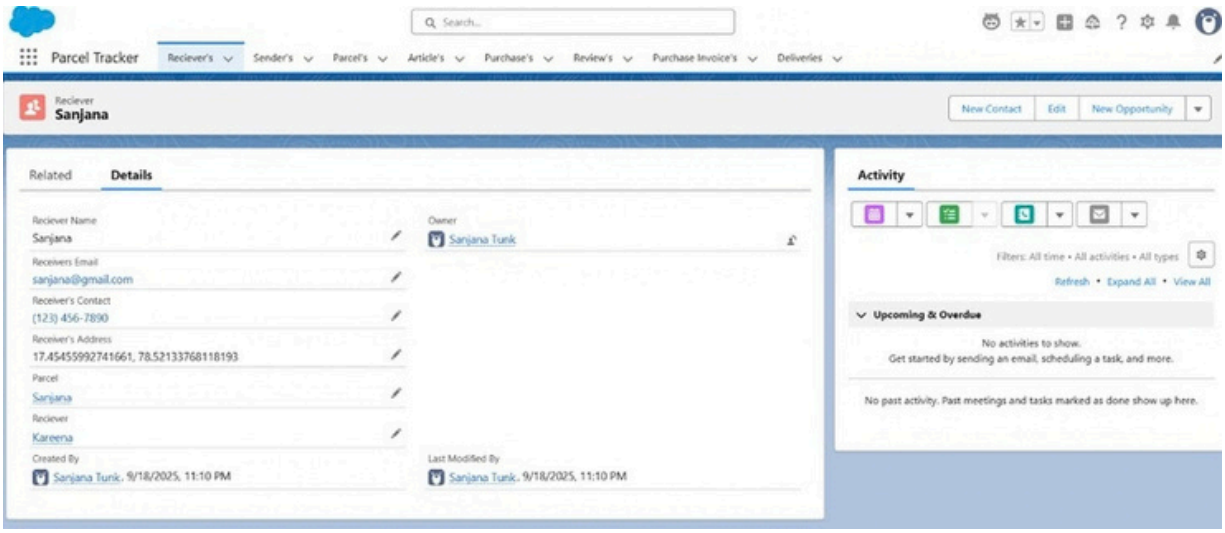



## Validate Data in Salesforce

1.After the agent runs the Flow:

- Go to App Launcher → Parcels (Parcel__c).
Search for the Parcel ID you used during testing.
Verify:

- The Status or related field is updated.
- Any Flow-created records (bookings, updates) appear correctly.




# Phase 5: Deployment, Documentation & Maintenance

● In the scope of our SwiftShip Tracker (AgentForce) project implementation:
● The purpose of this phase is to validate that the Parcel Tracking Agent, created using Salesforce Agent Builder, works as intended within the Salesforce Developer Org environment.
● The main goal is to understand how an AI-powered Service Agent can be tested to ensure correct topic recognition, Flow execution, and data retrieval from Salesforce custom objects.
● Testing is conducted within the Developer Org to simulate how the Agent would perform in a live setup, ensuring that permissions, flows, and interactions function correctly before any deployment stage.

● Real-world production testing (UAT or sandbox deployment) is not performed in this project because: Developer Edition orgs are standalone environments and do not support production-linked testing environments. Enterprise-level AgentForce deployment involves sandbox testing, staging environments, and CI/CD automation, which are outside the scope of this academic/prototype implementation.

## Maintenance, Monitoring & Troubleshooting

In the SwiftShip Tracker (AgentForce) project, ongoing maintenance ensures that Parcel, Customer, and Delivery data remain accurate, synchronized, and up to date. Regular monitoring helps detect issues such as misrouted agent queries, failed Flow executions, or performance delays in AgentForce responses. Troubleshooting activities include resolving configuration errors, updating Flow logic, refining agent topics, and improving prompts to enhance user interactions.
Although this project is implemented within a Salesforce Developer Org for learning purposes, in real-world enterprise deployments, these maintenance and monitoring practices are essential. They ensure that AI-powered service agents remain reliable, scalable, and continuously aligned with evolving customer service operations and business goals.

# Project Documentation

Project documentation in a Salesforce CRM implementation serves as a comprehensive record of the application's purpose, design, development, and deployment. It ensures that business requirements are clearly defined and aligned with the system’s functionality. Documentation acts as a blueprint for developers and admins, guiding consistent development and future scalability. It supports user training, facilitates troubleshooting, and ensures smooth maintenance by detailing every component and its logic. Moreover, it aids in change management, audit compliance, and knowledge transfer, making it an essential asset for the successful execution and long-term sustainability of any Salesforce CRM project.